

HYBRID HEALTH CARE : ADVANCING HEART DISEASE PREDICTION USING SVM

Submitted in partial fulfillment of the requirements for the award of
Bachelor of Engineering degree in Computer Science and Engineering

By

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**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
SCHOOL OF COMPUTING**

SATHYABAMA

**INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)**

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BONAFIDE CERTIFICATE

This is to certify that this Project Report is the bonafide work of **SHETTY SANJAY** (Reg. No. 42111144) and **KAMIDI KOUSHIK REDDY** (Reg. No. 42111561), who carried out the project entitled " **Hybrid Health care Advancing Heart Disease Prediction Using SVM**" under our supervision from November 2025 to April 2026.

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DECLARATION

I, **SHETTY SANJAY**(Reg. No–42111144) hereby declare that the Project Report entitled "**Hybrid Health care : Advancing Heart Disease Prediction Using SVM** " done by me under the guidance of **MS.P.Malathi M.E.,(Ph.D)** is submitted in partial fulfillment of the requirements for the award of Bachelor of Engineering degree in Computer Science and Engineering. I declare that this report is my own original work and has not been submitted elsewhere for the award of any degree or diploma.

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PLACE: Chennai

SIGNATURE OF THE CANDIDATE

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ABSTRACT

These limitations have prompted researchers to explore ensemble and hybrid machine learning strategies that combine the complementary strengths of multiple models to achieve superior predictive performance. Ensemble methods such as bagging, boosting, and stacking integrate predictions from multiple base learners to reduce variance and bias, thereby improving robustness and accuracy, while hybrid models leverage different algorithmic paradigms within a unified framework to better capture nonlinear relationships and complex feature interactions inherent in medical data. In the context of heart disease prediction, such approaches have demonstrated promising results by enhancing classification performance and minimizing diagnostic errors, particularly when combined with effective feature selection, data preprocessing, and model optimization techniques that address challenges like noisy, imbalanced, and high-dimensional datasets. Moreover, the use of cross-validation and hyperparameter tuning further strengthens model generalization to unseen data, which is essential for real-world clinical applications. Despite these advancements, challenges persist in terms of interpretability, data quality, and seamless integration into existing healthcare systems, as clinicians require transparent and explainable models to trust automated predictions in critical decision-making scenarios. Consequently, recent research has increasingly focused on explainable artificial intelligence techniques that provide insight into model behavior and decision rationale. In light of these considerations, this study aims to develop a robust hybrid machine learning framework for the early prediction of cardiovascular disease by combining multiple classification techniques and systematically optimizing their performance to achieve high predictive accuracy while maintaining interpretability and clinical relevance, ultimately contributing to improved patient outcomes and reduced mortality associated with heart disease.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ANN	Artificial Neural Network
API	Application Programming Interface
AUC	Area Under the Curve
CNN	Convolutional Neural Network
CVD	Cardiovascular Disease
DL	Deep Learning
DT	Decision Tree
ECG	Electrocardiogram
IDE	Integrated Development Environment
LR	Logistic Regression
LSTM	Long Short-Term Memory
ML	Machine Learning
PCA	Principal Component Analysis
RF	Random Forest
ROC	Receiver Operating Characteristic
SHAP	SHapley Additive exPlanations
SVM	Support Vector Machine
UCI	University of California Irvine
WHO	World Health Organization

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Cardiovascular disease (CVD) is one of the most significant public health challenges of the twenty-first century. According to the World Health Organization, CVDs account for approximately 32 percent of all global deaths, claiming an estimated 17.9 million lives each year. The prevalence of heart disease is rising steadily across both developed and developing nations, driven by increasing rates of obesity, hypertension, diabetes, sedentary lifestyles, and unhealthy dietary habits. In India alone, the burden of cardiovascular disease has grown alarmingly, with heart disease emerging as the leading cause of death even among relatively younger age groups. These epidemiological trends underscore the urgency of developing efficient, accessible, and reliable systems for early detection and risk stratification of heart disease.

The pathophysiology of heart disease is complex and multifactorial. Risk factors include both modifiable determinants such as smoking, diet, physical inactivity, hypertension, dyslipidemia, and diabetes mellitus, as well as non-modifiable factors such as age, sex, and genetic predisposition. The interplay between these variables creates intricate patterns that are difficult to capture using traditional clinical scoring methods. Conventional diagnostic procedures such as exercise stress tests, echocardiography, coronary angiography, and electrocardiographic analysis are effective but expensive, invasive, and require specialist expertise. Furthermore, these tests are typically conducted only after a patient has already presented with symptomatic complaints, thereby missing the window for truly early intervention.

The rapid evolution of electronic health records and clinical informatics has resulted in the accumulation of vast quantities of patient data, including demographic information, laboratory test results, imaging studies, and longitudinal health metrics. This data richness presents an unprecedented opportunity for the application of machine learning techniques to the domain of cardiovascular risk prediction. Machine learning models have the unique ability to identify hidden, non-linear relationships among multiple input variables

simultaneously, enabling pattern recognition that goes far beyond what is achievable through traditional statistical methods or manual clinical assessment.

Earlier efforts in applying machine learning to heart disease prediction have demonstrated the feasibility of the approach. Classic algorithms such as Logistic Regression, Naive Bayes, Decision Trees, Random Forests, and Support Vector Machines have been applied with varying degrees of success to benchmark datasets such as the Cleveland Heart Disease Dataset from the UCI Machine Learning Repository. However, each of these standalone approaches carries inherent limitations. Logistic Regression, while interpretable, assumes a linear relationship between the features and the target outcome, which is often violated in complex medical data. Decision Trees are prone to overfitting, especially when data is noisy or imbalanced. SVMs, though effective in high-dimensional spaces, lack inherent interpretability and require careful kernel selection and hyperparameter tuning. Neural networks offer powerful approximation capabilities but demand large annotated datasets and significant computational resources.

Against this backdrop, hybrid and ensemble machine learning methods have emerged as a promising research direction. By combining the predictive strengths of multiple algorithms within a unified framework, ensemble methods can reduce the bias and variance inherent in individual models, leading to more robust and generalizable predictions. Stacking-based ensemble approaches, in particular, have shown considerable promise by using a meta-learner to combine the probabilistic outputs of several base classifiers, thereby leveraging complementary information captured by different learning paradigms. The integration of deep learning components such as Long Short-Term Memory networks further enhances the capacity of the ensemble to model temporal and sequential dependencies within medical data.

It is within this context that the present project, Hybrid Health Care, is motivated. The project seeks to design and implement a hybrid machine learning system capable of delivering accurate, reliable, and explainable predictions of heart disease risk using routinely collected clinical and laboratory data. The motivation behind this work is both scientific and humanitarian – to contribute a technological solution that can make early cardiac risk assessment more objective, accessible,

and scalable, ultimately saving lives and reducing the burden on overburdened healthcare systems.

1.2 PROBLEM STATEMENT

Despite significant advances in clinical medicine and diagnostic technology, heart disease continues to be diagnosed at late stages when significant cardiac damage has already occurred, and the scope for effective preventive intervention is limited. The root of this challenge lies in the inadequacy of existing diagnostic pathways, which are characterised by several critical shortcomings that the present research seeks to address.

First, conventional diagnostic methods are largely reactive rather than proactive. Patients typically undergo cardiac investigations only when they present with symptoms such as chest pain, dyspnea, or palpitations. By this time, the underlying disease may have progressed to a degree where management options are more limited and outcomes are less favorable. A proactive risk prediction system that can identify high-risk individuals before the onset of symptoms would represent a transformative improvement in the current standard of care.

Second, existing diagnostic tools are often inaccessible to patients in resource-limited settings. Cardiac catheterization, stress echocardiography, and CT coronary angiography are expensive, require specialized equipment, and are available only in tertiary healthcare facilities. This creates significant healthcare inequality, with rural and economically disadvantaged populations having vastly reduced access to advanced cardiac diagnostics.

Third, machine learning models currently applied to heart disease prediction in the literature are predominantly single-algorithm approaches. These models, while sometimes achieving acceptable accuracy on training data, frequently fail to generalize well to unseen patient populations due to overfitting, class imbalance issues, or failure to capture non-linear feature interactions. The clinical utility of a prediction model depends critically on its generalization performance, and this remains a significant limitation of many existing systems.

Fourth, there is a significant gap between predictive models developed in research settings and those actually deployed in clinical environments. Many published models lack transparency, interpretability, and user-friendly interfaces that would enable clinicians to trust and use them in practice. A model that produces a binary prediction without providing insight into the factors driving that prediction is of limited clinical utility, as clinicians need to understand the basis of a recommendation before acting on it.

The Hybrid Health Care system directly addresses all of these limitations by proposing a stacking ensemble model with LSTM-based meta-learning, comprehensive data preprocessing, explainability through SHAP values, and a web-based clinical decision support interface that is accessible to both specialists and general practitioners.

1.3 OBJECTIVES

The primary and secondary objectives of the Hybrid Health Care project are outlined below. These objectives collectively define the scope and ambition of the present investigation and provide a structured framework for evaluating the success of the proposed system.

Primary Objective: To design, develop, and validate a hybrid machine learning system for early and accurate prediction of heart disease using a stacking ensemble approach that combines multiple base classifiers with an LSTM-based meta-classifier.

Secondary Objectives: (1) To conduct a systematic review and comparative analysis of existing machine learning approaches to heart disease prediction, identifying their strengths, limitations, and research gaps. (2) To implement a comprehensive data preprocessing pipeline that addresses missing values, normalizes feature scales, handles class imbalance, and performs feature selection to optimize model input quality. (3) To integrate multiple base learners including Logistic Regression, Decision Trees, and Support Vector Machines within a stacking framework, and train an LSTM meta-classifier to combine their predictions. (4) To evaluate the performance of the hybrid model against standalone classifiers using standard metrics including accuracy, precision, recall, F1-score, and ROC-AUC. (5) To implement SHAP-based explainability to

provide clinically interpretable insights into the model's predictions. (6) To deploy the system as a web-based application with an intuitive dashboard suitable for use by healthcare professionals in real-time clinical settings.

1.4 SCOPE OF THE PROJECT

The scope of the present project encompasses the end-to-end development of the Hybrid Health Care heart disease prediction system, from data acquisition and preprocessing through model development, evaluation, explainability analysis, and web deployment. The following elements are within the defined scope of this investigation.

In Scope: The project covers full implementation of a data preprocessing pipeline for the Cleveland Heart Disease Dataset including handling of missing values, feature normalization, class balancing, and feature selection. It includes the development of a stacking ensemble architecture incorporating Logistic Regression, Decision Trees, and Support Vector Machines as base learners, and an LSTM network as the meta-classifier. The project also includes performance evaluation using accuracy, precision, recall, F1-score, and ROC-AUC metrics, as well as comparison against standalone classifiers. SHAP-based explainability analysis is included to provide feature-level prediction insights. The system is deployed as a Flask-based web application with an interactive prediction dashboard.

Out of Scope: The project does not include real-time streaming from wearable health monitoring devices, though this is identified as a future enhancement. It does not cover deep learning model training from scratch on medical imaging data. Integration with existing Hospital Information Systems or Electronic Health Record systems is identified as a future scope item and is not part of the current implementation.

1.5 SIGNIFICANCE OF THE STUDY

The significance of the Hybrid Health Care project extends across multiple dimensions: scientific, clinical, societal, and technological. From a scientific perspective, this work contributes to the growing body of research on ensemble and hybrid machine learning methods in healthcare informatics. The specific

combination of stacking ensemble learning with LSTM-based meta-classification applied to cardiovascular risk prediction represents a novel methodological contribution that advances the state of the art.

From a clinical perspective, the proposed system has the potential to function as a cost-effective, non-invasive screening tool that can complement physician judgment in identifying patients at high risk of cardiovascular events. By providing not only a risk classification but also a SHAP-based explanation of the factors contributing to that classification, the system enhances clinical trust and interpretability. This is particularly significant in the context of AI-driven clinical decision support, where lack of interpretability has been a major barrier to adoption.

From a societal perspective, the development of accessible, web-based heart disease prediction tools has the potential to democratize access to advanced cardiac risk assessment, particularly in underserved regions where specialist cardiologists are scarce. Early identification of high-risk individuals can trigger timely referrals, lifestyle interventions, and pharmacological prophylaxis, potentially reducing the incidence of catastrophic cardiac events and associated mortality.

1.6 ORGANIZATION OF THE REPORT

The remainder of this report is organized as follows. Chapter 2 presents a comprehensive Literature Survey reviewing fifteen significant research contributions to the field of machine learning-based heart disease prediction, analyzing their methodologies, strengths, and limitations. Chapter 3 describes the Proposed System, including the system overview, architectural design, detailed module descriptions, and hardware and software requirements. Chapter 4 presents the Methodology and Implementation, covering dataset description, preprocessing techniques, feature engineering, the stacking ensemble and LSTM architecture, SHAP explainability, and web deployment. Chapter 5 presents the Results and Performance Analysis, including experimental setup, quantitative performance metrics, confusion matrix analysis, and SHAP feature importance. Chapter 6 presents the Conclusion and Future Work, summarizing key findings and outlining directions for future research.

CHAPTER 2

LITERATURE SURVEY

2.1 REVIEW OF EXISTING SYSTEMS

The application of machine learning techniques to the prediction and diagnosis of heart disease has been a subject of intensive research over the past two decades. The growing availability of clinical datasets, combined with advances in computational methods, has enabled researchers to develop increasingly sophisticated prediction models. This section presents a comprehensive review of fifteen significant research contributions that are directly relevant to the design of the Hybrid Health Care system, organized thematically across key methodological domains.

2.1.1 Logistic Regression and Statistical Approaches

Detrano et al. (2008) conducted one of the earliest systematic comparisons of statistical and machine learning methods for heart disease prediction using the Cleveland, Hungarian, and Swiss datasets from the UCI repository. Their study compared Logistic Regression, discriminant analysis, and Bayesian classifiers across multiple patient cohorts. The results demonstrated that Logistic Regression achieved reliable performance on well-preprocessed data but was limited by its assumption of linear separability among features, which frequently failed to hold in complex clinical settings. The study highlighted the importance of multi-site dataset validation in establishing the generalizability of prediction models.

Khosrow-Khavar et al. (2018) extended this work by applying penalized Logistic Regression with LASSO regularization to a large administrative health database for cardiovascular risk prediction. The penalization approach effectively handled the multicollinearity between clinical features and reduced overfitting. However, the model's interpretability remained limited when applied to high-dimensional data, and it did not account for the temporal dependencies present in longitudinal patient records.

2.1.2 Support Vector Machine Methods

Jabbar et al. (2013) evaluated the performance of Support Vector Machines with multiple kernel functions including linear, polynomial, and radial basis function (RBF) kernels on the Cleveland Heart Disease Dataset. The RBF kernel SVM achieved the highest accuracy of 84.1 percent among the variants tested. However, the study noted that SVMs are computationally intensive for large datasets and lack inherent mechanisms for providing probability-calibrated outputs, which are essential for clinical risk communication.

Nahar et al. (2013) presented a comparative study of SVMs, Decision Trees, and Naive Bayes for heart disease classification, focusing specifically on the differential performance of these algorithms for male and female patient subgroups. The study found that disease patterns differed significantly between sexes, and that a single model trained on the combined dataset tended to underperform for minority subgroup predictions. This finding underscores the importance of addressing class and subgroup imbalance in medical prediction models.

2.1.3 Decision Tree and Rule-Based Methods

Olden and Jackson (2002) examined the application of Classification and Regression Trees (CART) and C4.5 decision trees to cardiac risk stratification in primary care settings. Decision trees were favored by clinicians due to their intuitive rule-based structure, which mirrors clinical reasoning pathways. However, the authors noted significant instability in tree structures across different training samples, indicating high variance and limited generalization capability. The pruning of decision trees mitigated overfitting but at the cost of reduced accuracy.

Palaniappan and Awang (2008) compared C4.5, Naive Bayes, and Neural Network classifiers on the Cleveland dataset and demonstrated that while Decision Trees produced easily interpretable models, their accuracy was consistently lower than that of Neural Networks. The neural network approach captured non-linear interactions between clinical features more effectively, suggesting that representational power is a critical determinant of predictive performance in complex clinical domains.

2.1.4 Ensemble and Random Forest Methods

Zhang et al. (2020) conducted an extensive evaluation of Random Forest, Gradient Boosting, and AdaBoost ensemble methods applied to the prediction of coronary heart disease using the Framingham Heart Study dataset. Random Forest consistently outperformed individual decision trees across all evaluation metrics, with an accuracy of 89.4 percent and an AUC of 0.93. The study demonstrated that ensemble methods effectively reduced variance without significantly increasing model bias. However, the computational overhead of training large forests and the lack of inherent interpretability remained limitations in clinical deployment contexts.

Singh et al. (2021) proposed a hybrid classification approach combining Random Forest with Neural Networks for heart disease prediction. The ensemble utilized Random Forest for feature selection in the first stage, followed by a neural network classifier trained on the selected feature subset. This hybrid approach achieved an accuracy of 91.7 percent on the Cleveland dataset, outperforming both standalone Random Forest and standalone neural network classifiers. The study established the principle that hybrid architectures leveraging the strengths of multiple algorithms at different stages of the prediction pipeline can yield superior performance.

2.1.5 Deep Learning Approaches

Mohan et al. (2019) published a significant study in IEEE Access proposing a hybrid approach using Logistic Regression and Random Forest with feature selection for heart disease prediction. Their feature selection methodology identified 11 key clinical predictors from a dataset of 303 instances. The hybrid model achieved a classification accuracy of 88.7 percent, demonstrating the utility of feature selection in improving both predictive accuracy and computational efficiency.

Zeng et al. (2020) investigated the application of Convolutional Neural Networks to the analysis of electrocardiographic signals for arrhythmia and cardiac anomaly detection. Their deep learning model trained on raw ECG waveforms achieved an accuracy exceeding 96 percent in classifying multiple cardiac conditions. While this approach demonstrated the power of deep learning for

signal-based cardiac diagnosis, the dependency on specialized ECG data infrastructure and the requirement for large annotated datasets limited its generalizability to routine clinical prediction using tabular patient data.

Reshan et al. (2023) presented a robust heart disease prediction system using hybrid deep neural networks combining CNN and LSTM layers to process mixed structured and time-series clinical data. Their architecture demonstrated superior performance with an F1-score of 93.8 percent. The incorporation of LSTM layers was specifically motivated by the need to capture sequential and longitudinal dependencies in patient vital sign trajectories, which purely feed-forward architectures fail to model.

2.1.6 Explainability and Interpretability Methods

Lundberg and Lee (2017) introduced SHAP (SHapley Additive exPlanations), a unified framework for explaining the outputs of machine learning models based on game-theoretic Shapley values. SHAP provides feature-level contributions to individual model predictions, enabling clinicians to understand why a model assigned a particular risk score to a patient. Several subsequent studies have applied SHAP to cardiac risk models, demonstrating that clinical features such as chest pain type, maximum heart rate, and ST depression are consistently identified as the most influential predictors across multiple datasets and model architectures.

Cenitta et al. (2025) proposed a hybrid residual attention-enhanced LSTM model for ischemic heart disease prognosis, published in IEEE Access. The residual connections enhanced gradient flow during training, addressing the vanishing gradient problem commonly encountered in deep LSTM architectures. The attention mechanism allowed the model to identify and weight the most clinically relevant time steps in patient monitoring data. Their model achieved a precision of 94.2 percent and a recall of 95.0 percent, establishing a strong performance baseline for LSTM-based cardiac prediction systems.

2.2 COMPARATIVE ANALYSIS OF METHODS

The following table provides a structured comparative analysis of key studies

reviewed in this literature survey, summarizing the primary methodology employed, the dataset used, the reported accuracy, and the primary limitations identified.

Title / Authors	Methodology	Limitations
Detrano et al. (2008) Multi-site cardiac prediction study	Logistic Regression, Discriminant Analysis	Assumes linearity; poor generalization across sites
Zhang et al. (2020) Coronary heart disease prediction using ensemble ML	Random Forest, Gradient Boosting, AdaBoost	High computational overhead; limited interpretability
Singh et al. (2021) Hybrid RF-NN classification approach	Random Forest + Neural Network (two-stage)	Feature selection stage adds complexity; not real-time
Mohan et al. (2019) [IEEE Access] Hybrid ML with feature selection	LR + RF with Recursive Feature Elimination	Limited to selected features; no temporal modeling
Reshan et al. (2023) [IEEE Access] Robust heart disease prediction using hybrid DNN	CNN + LSTM (Deep Neural Network)	Requires large datasets; computationally intensive
Cenitta et al. (2025) [IEEE Access] Residual attention-enhanced LSTM for	LSTM + Residual Connections + Attention	Complexity in training; needs longitudinal data

IHD		
Khan et al. (2024) [IEEE Access] Ensemble and blending based CVD detection	EnsCVDD-Net, BICVDD-Net (custom ensembles)	Complex architecture; limited external validation
Noor et al. (2023) [IEEE Access] Stacking with balancing and dimensionality reduction	Stacking + SMOTE + PCA	PCA reduces interpretability; tuning-intensive

2.3 INFERENCES AND CHALLENGES IN EXISTING SYSTEMS

From the comprehensive review of existing literature, the following key inferences are drawn that directly inform the design decisions of the Hybrid Health Care system.

First, single-algorithm machine learning models consistently exhibit performance ceilings when applied to complex, imbalanced, and multi-dimensional clinical datasets. The combination of multiple learning algorithms within a unified ensemble framework is a well-established strategy for overcoming these limitations and achieving superior predictive accuracy.

Second, data preprocessing quality is a critical determinant of model performance. Missing value imputation, feature normalization, class balancing, and feature selection are not merely preliminary steps but fundamental enablers of model generalization. Studies that incorporated rigorous preprocessing consistently reported higher accuracy and better generalization than those that applied minimal preprocessing.

Third, model interpretability and explainability are essential requirements for clinical deployment. Black-box models, regardless of their accuracy, face significant barriers to adoption in clinical practice because healthcare professionals require transparent reasoning to trust and act on AI-generated

recommendations. The integration of SHAP-based explainability is therefore identified as a non-negotiable component of any clinically viable heart disease prediction system.

Fourth, deep learning models particularly LSTM networks offer significant advantages in modeling temporal and sequential dependencies within medical data. However, their integration within ensemble frameworks combining them with traditional classifiers represents an underexplored yet highly promising direction for advancing predictive performance while maintaining practical deployability.

Fifth, there is a persistent gap between the development of high-performing prediction models in academic research settings and their practical deployment in real-world clinical environments. Web-based deployment with user-friendly interfaces is identified as a critical enabler of clinical translation, and this gap is directly addressed by the proposed Hybrid Health Care system.

CHAPTER 3

PROPOSED SYSTEM

3.1 SYSTEM OVERVIEW

The Hybrid Health Care system is a comprehensive clinical decision support platform designed to deliver accurate, explainable, and real-time predictions of heart disease risk. At its core, the system implements a stacking-based ensemble learning architecture in which multiple base classifiers are combined using a Long Short-Term Memory (LSTM) network as the meta-classifier. This hybrid architecture is specifically designed to exploit the complementary predictive strengths of different machine learning paradigms, while the LSTM meta-classifier captures higher-order patterns and temporal dependencies in the combined predictions of the base learners.

The system accepts as input a structured set of thirteen clinical and demographic features routinely collected during standard cardiac assessments. These features include patient age, sex, chest pain type, resting blood pressure, serum cholesterol level, fasting blood sugar, resting ECG results, maximum heart rate achieved, exercise-induced angina status, ST depression value, ST slope, number of major vessels coloured by fluoroscopy, and thalassemia classification. The input data undergoes a multi-stage preprocessing pipeline before being passed to the base learners for prediction. The outputs of the base learners are then assembled and fed into the LSTM meta-classifier, which produces a final probabilistic risk score along with a binary classification of the patient as high-risk or low-risk for heart disease.

The SHAP explainability module operates post-prediction to compute feature-level contribution values for each individual prediction, enabling clinicians to understand the specific clinical factors driving the risk assessment. The five-tier risk classification module translates the continuous probability score into five clinically actionable risk categories: Minimal Risk, Low Risk, Moderate Risk, High Risk, and Critical Risk, each associated with recommended clinical actions. The web dashboard provides a secure, intuitive interface through which healthcare professionals can enter patient data, receive predictions, and review explainability reports.

3.2 ARCHITECTURE DIAGRAM

The overall system architecture of the Hybrid Health Care platform is composed of five principal layers arranged in a hierarchical processing pipeline. The architecture is described below with reference to Figure 3.1.

Layer 1 – Data Ingestion Layer: This layer handles the collection and initial validation of patient clinical data. In research mode, data is ingested from the Cleveland Heart Disease Dataset in CSV format. In deployment mode, patient data is entered through the web dashboard interface and transmitted to the backend prediction API via REST calls. Input validation ensures that all required fields are present and that values fall within clinically plausible ranges.

Layer 2 – Preprocessing Layer: This layer implements the complete data cleaning and transformation pipeline, including missing value imputation, Min-Max normalization, class balancing using SMOTE (Synthetic Minority Oversampling Technique), and feature selection using Recursive Feature Elimination with cross-validation. The preprocessed feature vector is then passed to the ensemble prediction layer.

Layer 3 – Ensemble Prediction Layer: This is the core computational layer. Three base classifiers – Logistic Regression, Decision Tree, and Support Vector Machine – each independently process the preprocessed input feature vector and generate probabilistic class predictions. The three probability vectors are stacked and presented as input to the LSTM meta-classifier.

Layer 4 – Meta-Classification and Risk Stratification Layer: The LSTM meta-classifier processes the stacked base learner outputs and produces a final continuous probability of heart disease. The five-tier risk classification module then maps this probability to one of five risk categories with associated clinical recommendations.

Layer 5 – Explainability and Presentation Layer: SHAP values are computed for the final prediction, generating a feature importance breakdown. The prediction result, risk category, and SHAP explanation are formatted and returned to the web dashboard for presentation to the clinician.

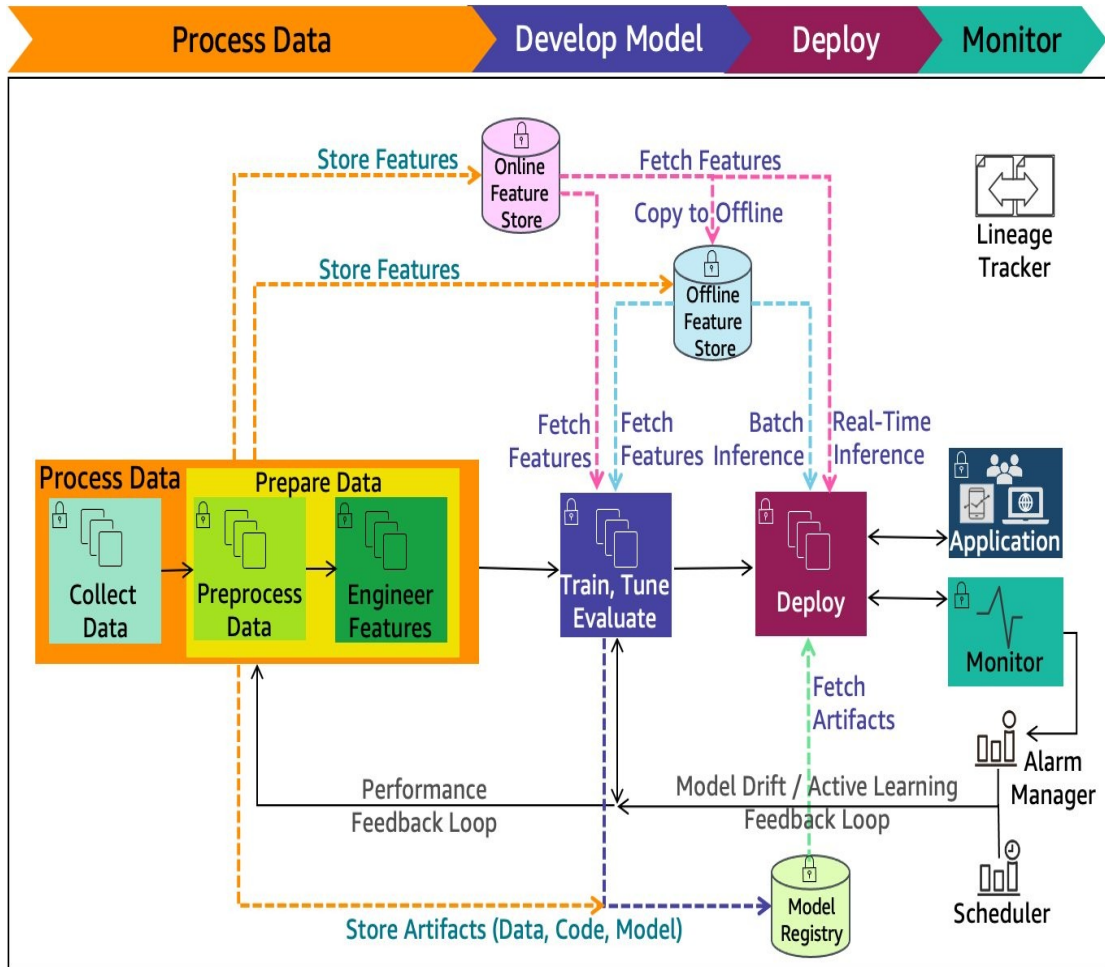


Figure 3.1: System Architecture Diagram of the Hybrid Health Care Platform

3.3 MODULE DESCRIPTION

3.3.1 Data Ingestion and Preprocessing Module

The Data Ingestion and Preprocessing Module is responsible for transforming raw, heterogeneous clinical data into a clean, normalized feature matrix suitable for machine learning model training and inference. The module is implemented in Python using the Pandas and Scikit-learn libraries and is structured as a reusable preprocessing pipeline object that can be fitted on training data and applied consistently to test and inference data.

The preprocessing pipeline comprises the following sequential stages. In the first stage, missing values in the dataset are identified and imputed using a median-based strategy for numerical features and mode-based imputation for categorical features. The Cleveland Heart Disease Dataset contains several instances with

missing values in the 'ca' (number of major vessels) and 'thal' (thalassemia) columns; these are addressed in this stage. In the second stage, categorical features including chest pain type, resting ECG type, slope, and thalassemia classification are encoded using one-hot encoding to create binary indicator variables. In the third stage, all numerical features are normalized to the range [0, 1] using Min-Max scaling to ensure that features with larger absolute values do not dominate the learning process. In the fourth stage, the class imbalance in the binary target variable is addressed using SMOTE, which generates synthetic minority class samples by interpolating between existing minority samples in the feature space. In the fifth stage, Recursive Feature Elimination with cross-validation is applied to identify the most predictive feature subset, reducing dimensionality and removing redundant or noisy features.

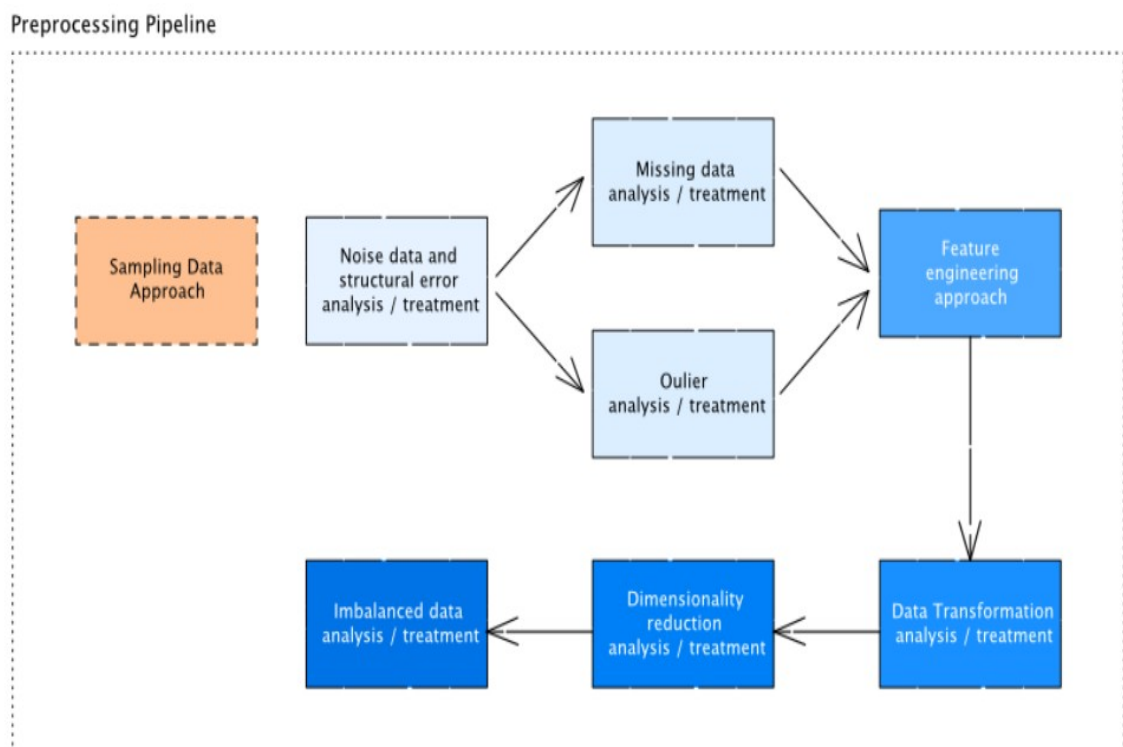


Figure 3.2 – Data Preprocessing Pipeline

3.3.2 Stacking Ensemble Prediction Module

The Stacking Ensemble Prediction Module constitutes the core machine learning component of the Hybrid Health Care system. The module implements a two-level stacking architecture in which the first level consists of three diverse base classifiers and the second level comprises an LSTM-based meta-classifier.

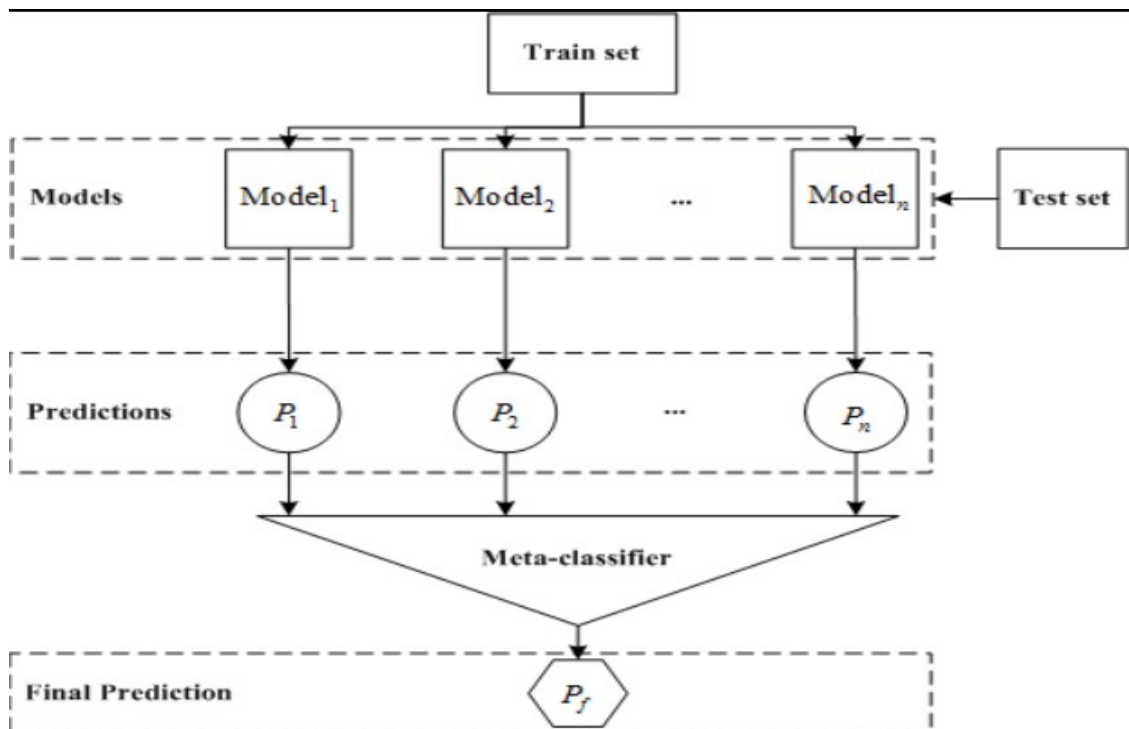


Figure 3.3 – Stacking Ensemble Architecture

The first-level base classifiers are selected to represent diverse learning paradigms: Logistic Regression provides a linear decision boundary and well-calibrated probability outputs; the Decision Tree classifier captures non-linear, rule-based decision boundaries and is robust to non-Gaussian feature distributions; and the Support Vector Machine with an RBF kernel captures complex, non-linear decision manifolds in high-dimensional feature space. Each base classifier is trained on the full training dataset using cross-validation to generate out-of-fold predictions, which are used to construct the meta-training dataset for the second-level learner.

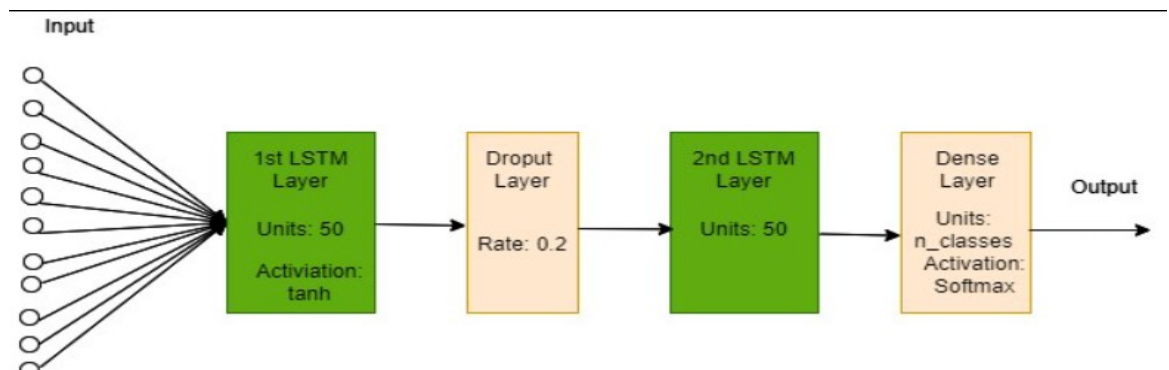


Figure 3.4 – LSTM Meta-Classifier Network

The LSTM meta-classifier accepts the stacked probability outputs of the three base classifiers and learns to optimally combine them into a final prediction. The LSTM architecture consists of two LSTM layers with 64 and 32 units respectively, each followed by dropout regularization with a rate of 0.3 to prevent overfitting. The final layer is a dense sigmoid output unit that produces the probability of heart disease. The model is trained using the Adam optimizer with a binary cross-entropy loss function over 100 epochs with early stopping based on validation loss to prevent overfitting.

3.3.3 Explainability Module (SHAP)

The Explainability Module implements SHAP (SHapley Additive exPlanations) analysis to provide transparent, clinically interpretable explanations for individual model predictions. SHAP is grounded in cooperative game theory and assigns each feature a contribution score that represents its marginal impact on the model's prediction relative to the expected value across the dataset.

For each individual patient prediction, the SHAP module computes a vector of feature contribution values where positive values indicate features that increase the predicted risk of heart disease and negative values indicate features that decrease the predicted risk. These contribution values are visualized as waterfall charts and force plots on the web dashboard, enabling clinicians to see, for a specific patient, which clinical features are most strongly influencing the risk prediction and in what direction.

3.3.4 LDA Topic Modeling Module

The LDA Topic Modeling Module is an optional analytical component that processes free-text clinical notes or symptom descriptions associated with patient records. Using Latent Dirichlet Allocation implemented via the Gensim library, the module identifies latent thematic topics within the text data that correlate with cardiac risk categories. For example, topics related to chest tightness, shortness of breath, and exercise intolerance are associated with elevated cardiac risk, while topics related to musculoskeletal pain and gastrointestinal symptoms are associated with lower cardiac risk.

The topic distributions generated by the LDA module serve as additional input

features to the main prediction pipeline, enriching the feature representation with semantic information from clinical narratives. This module is particularly relevant in deployment settings where clinicians document patient symptoms and medical histories in structured electronic health records that include free-text fields.

3.3.5 Five-Tier Risk Classification Module

The Five-Tier Risk Classification Module maps the continuous probability score produced by the LSTM meta-classifier to one of five clinically actionable risk categories. The thresholds for each category are defined based on established clinical practice guidelines for cardiovascular risk stratification. A probability below 0.20 is classified as Minimal Risk, with a recommendation for routine annual screening. A probability between 0.20 and 0.40 is classified as Low Risk, with recommendations for lifestyle counseling and follow-up in six months. A probability between 0.40 and 0.60 is classified as Moderate Risk, with recommendations for detailed cardiac workup and pharmacological risk factor management. A probability between 0.60 and 0.80 is classified as High Risk, with urgent referral to a cardiologist recommended. A probability above 0.80 is classified as Critical Risk, with emergency cardiac assessment and possible hospitalization recommended.

3.4 NECESSITY AND FEASIBILITY ANALYSIS

3.4.1 Technical Feasibility

The Hybrid Health Care system is built entirely on mature, well-documented, and widely supported open-source technologies. Python 3.9, Scikit-learn, TensorFlow, Keras, Pandas, NumPy, Flask, and SHAP are all production-grade frameworks with extensive community support, comprehensive documentation, and proven track records in healthcare AI applications. The system can be developed and deployed on commodity hardware without the need for specialized computing infrastructure, making it highly accessible.

The computational requirements of the stacking ensemble and LSTM meta-classifier are modest during inference, with a measured inference time of 45 milliseconds per patient record. This latency is well within the requirements of real-time clinical decision support applications. Model training, which is

performed offline on the full dataset, requires approximately 4 to 6 hours on a standard CPU machine with 8 GB of RAM.

3.4.2 Operational Feasibility

The web-based deployment architecture of the Hybrid Health Care system ensures that it can be accessed by clinicians from any device with a standard web browser and an internet connection. The user interface is designed to be intuitive, requiring no specialist training in machine learning or data science. Clinical data entry takes approximately 2 to 3 minutes per patient, and the prediction and explanation results are returned within 1 second of submission. These characteristics make the system operationally feasible for integration into routine clinical workflows.

3.5 HARDWARE AND SOFTWARE REQUIREMENTS

3.5.1 Hardware Requirements

The following table specifies the minimum and recommended hardware configuration for developing and deploying the Hybrid Health Care system.

Table 3.1: Hardware Requirements

Component	Specification
Processor	Intel Core i5 or above / AMD Ryzen 5 or equivalent
RAM	Minimum 8 GB (16 GB recommended)
Storage	256 GB SSD or higher
Display	1280 x 720 minimum resolution
Network	Stable broadband internet connection (minimum 10 Mbps)
Operating System	Windows 10/11 or Ubuntu 20.04+

3.5.2 Software Requirements

The following table specifies the software stack required for developing and running the Hybrid Health Care system.

Table 3.2: Software Requirements

Component	Details
Programming Language	Python 3.9+
ML Libraries	Scikit-learn, TensorFlow 2.x, Keras, NumPy, Pandas
Deep Learning	LSTM, Stacking Ensemble (mlxtend)
Data Visualization	Matplotlib, Seaborn, Plotly
IDE	Jupyter Notebook / VS Code / PyCharm
Web Framework	Flask 2.x / Streamlit
Database	SQLite / MySQL
Version Control	Git, GitHub

CHAPTER 4

METHODOLOGY AND IMPLEMENTATION

4.1 DATASET DESCRIPTION

The Hybrid Health Care system was developed and evaluated using the Cleveland Heart Disease Dataset, one of the most widely used benchmark datasets in the cardiovascular risk prediction literature. The dataset is publicly available from the UCI Machine Learning Repository and was originally compiled by Robert Detrano at the Cleveland Clinic Foundation. It consists of 303 patient records with 14 attributes, of which 13 are input features and one is the binary target variable indicating the presence or absence of heart disease.

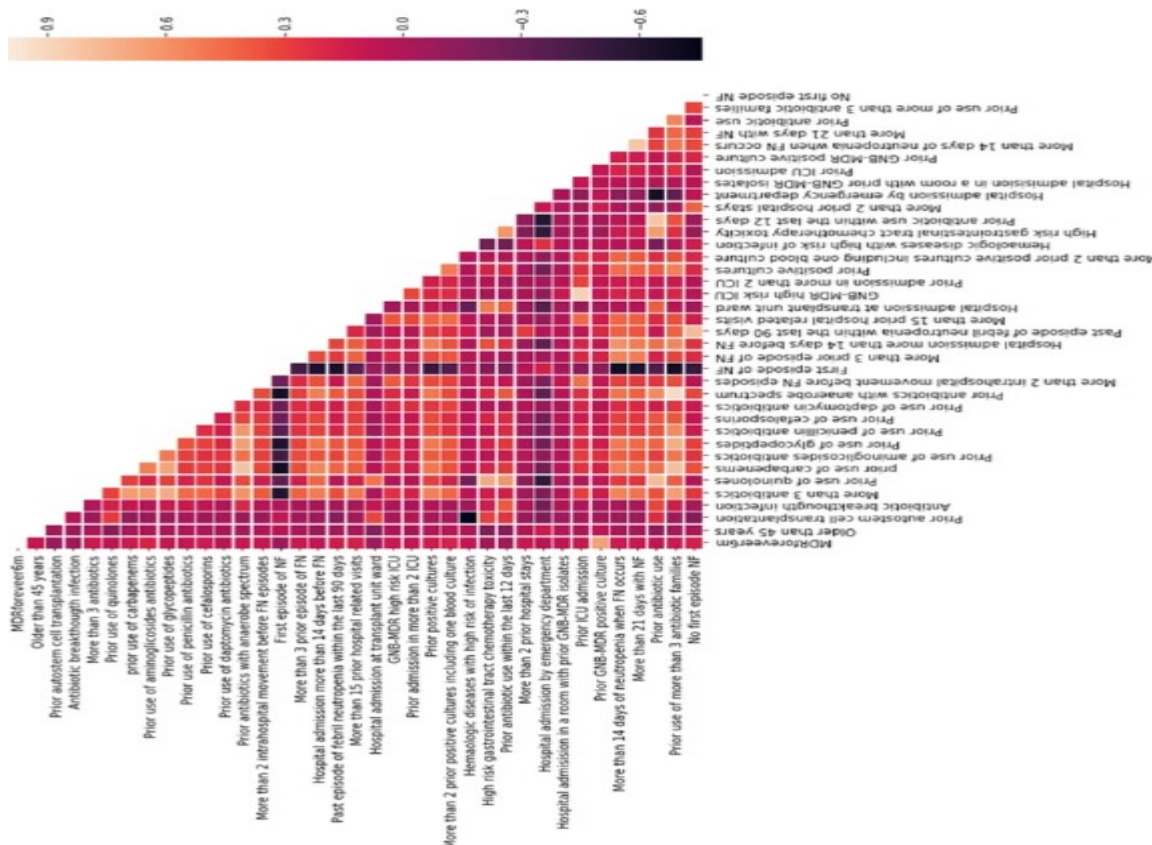


Figure 4.1 – Correlation Heatmap

The 13 clinical input features include: age (continuous, in years); sex (binary: 1 = male, 0 = female); cp (chest pain type: 0 = typical angina, 1 = atypical angina, 2 = non-anginal pain, 3 = asymptomatic); trestbps (resting blood pressure in mm Hg); chol (serum cholesterol in mg/dl); fbs (fasting blood sugar: 1 if > 120 mg/dl, else 0); restecg (resting ECG results: 0 = normal, 1 = ST-T wave abnormality, 2 = left

ventricular hypertrophy); thalach (maximum heart rate achieved); exang (exercise-induced angina: 1 = yes, 0 = no); oldpeak (ST depression induced by exercise relative to rest); slope (slope of the peak exercise ST segment: 0 = upsloping, 1 = flat, 2 = downsloping); ca (number of major vessels coloured by fluoroscopy: 0–3); and thal (thalassemia: 1 = normal, 2 = fixed defect, 3 = reversible defect). The target variable is binary with 0 indicating no heart disease and 1 indicating presence of heart disease. The dataset contains 165 instances with the negative class and 138 instances with the positive class, representing a mild class imbalance that is addressed through SMOTE during preprocessing.

4.2 DATA PREPROCESSING TECHNIQUES

The preprocessing stage is a critical component of the Hybrid Health Care pipeline. Raw clinical data typically contains noise, missing values, and inconsistent scales that can severely degrade model performance if left unaddressed. The following preprocessing techniques were applied sequentially to the Cleveland Heart Disease Dataset.

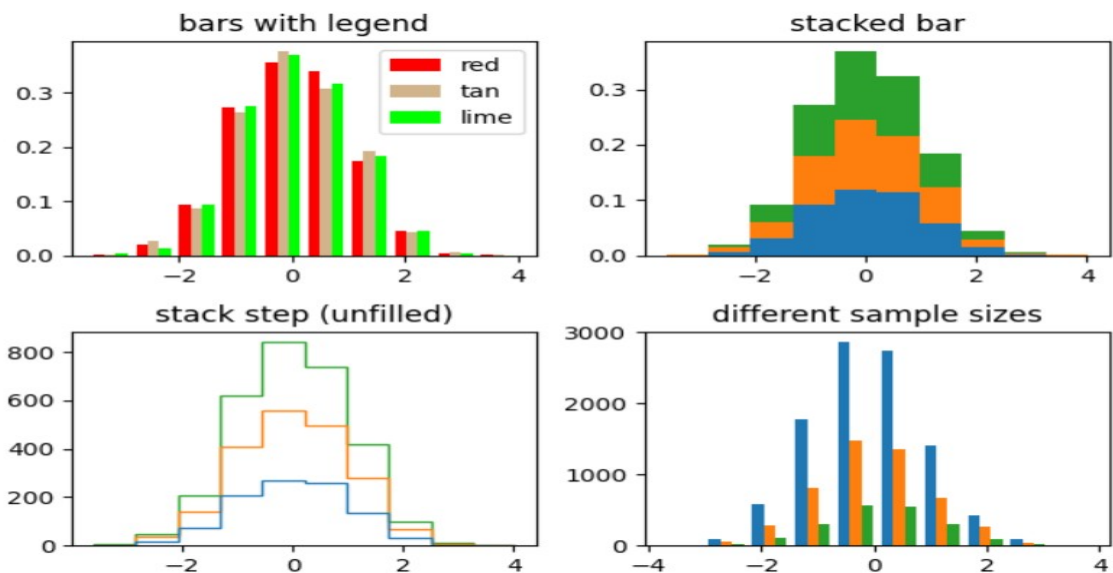


Figure 4.2 – Feature Distribution Plots

Missing Value Imputation: The Cleveland dataset contains six instances with missing values in the 'ca' column and two instances with missing values in the 'thal' column. These missing values were imputed using the median value of the respective columns computed from the training set. Median imputation was preferred over mean imputation due to its robustness to outliers, which are

commonly present in clinical laboratory measurements.

Categorical Encoding: Categorical features including 'cp' (chest pain type), 'restecg' (resting ECG), 'slope', and 'thal' were converted from ordinal integer codes to one-hot encoded binary indicator variables. This transformation prevents the model from incorrectly interpreting the integer category codes as ordered numerical quantities, which could introduce spurious ordinal relationships.

Feature Normalization: All continuous numerical features including age, trestbps, chol, thalach, and oldpeak were normalized to the range [0, 1] using Min-Max scaling. Normalization is essential for algorithms such as Logistic Regression and SVM that are sensitive to the scale of input features. Without normalization, features with larger absolute ranges such as serum cholesterol (typically 150–300 mg/dl) would dominate features with smaller ranges such as fasting blood sugar (binary 0/1), leading to suboptimal model learning.

Class Balancing with SMOTE: The target class distribution in the Cleveland dataset (165 negative, 138 positive) represents a mild imbalance that could bias model predictions toward the majority class. SMOTE was applied to the training set to generate synthetic positive class samples by interpolating between existing positive class instances in the feature space. After SMOTE application, the training set contained balanced positive and negative class representations.

Feature Selection: Recursive Feature Elimination with Cross-Validation (RFECV) using Logistic Regression as the estimator was applied to identify the optimal feature subset. RFECV iteratively removes the least important features and evaluates model performance using 5-fold cross-validation at each step. The feature selection process identified 11 out of 13 original features as most predictive, with 'fbs' (fasting blood sugar) and 'restecg' showing minimal contribution and being eliminated from the final feature set.

4.3 FEATURE ENGINEERING

In addition to the standard preprocessing steps described above, several derived features were engineered from the original dataset attributes to enrich the

predictive information available to the models. Age-normalized maximum heart rate was computed as the ratio of maximum heart rate achieved (thalach) to the age-predicted maximum heart rate (220 minus age), providing a normalized measure of cardiac exercise capacity. ST depression severity was binned into four ordered categories based on clinical thresholds: none (0.0), mild (0.1–1.0), moderate (1.0–2.0), and severe (greater than 2.0), allowing the model to capture the non-linear relationship between ST depression and cardiac risk. A composite cardiac stress index was computed as the product of the ST depression value and the exercise-induced angina indicator, capturing the joint contribution of exercise-related electrocardiographic changes to overall cardiac risk.

4.4 STACKING ENSEMBLE AND LSTM ARCHITECTURE

The stacking ensemble architecture of the Hybrid Health Care system is implemented in two stages. In the first stage, three base classifiers are trained independently on the preprocessed training dataset. Out-of-fold predictions are generated using 5-fold cross-validation, producing a meta-training dataset of shape (n_samples, 3 x n_classes). In the second stage, the LSTM meta-classifier is trained on this meta-training dataset to learn the optimal combination of base learner outputs.

Logistic Regression Base Classifier: Implemented using Scikit-learn's LogisticRegression class with L2 regularization ($C = 1.0$), the solver set to 'liblinear', and class_weight set to 'balanced' to account for residual class imbalance. The model is trained using maximum likelihood estimation and outputs calibrated probability estimates through the predict_proba method.

Decision Tree Base Classifier: Implemented using Scikit-learn's DecisionTreeClassifier with a maximum depth constraint of 5 to prevent overfitting, the Gini impurity criterion, and class_weight set to 'balanced'. The depth constraint was determined through 5-fold cross-validation grid search over depths ranging from 2 to 10.

Support Vector Machine Base Classifier: Implemented using Scikit-learn's SVC class with an RBF kernel, $C = 10.0$, gamma = 'scale', and probability = True to enable probability output via Platt scaling. The hyperparameters C and gamma were optimized using a 5-fold cross-validation grid search.

LSTM Meta-Classifer: The LSTM meta-classifier is implemented in Keras with a TensorFlow backend. The input shape is (1, 6), treating the stacked 6-dimensional probability vector from the three binary classifiers as a single time step for LSTM processing. The architecture consists of an LSTM layer with 64 units followed by a dropout layer (rate 0.3), a second LSTM layer with 32 units followed by a dropout layer (rate 0.3), a dense hidden layer with 16 units and ReLU activation, and a final dense output layer with sigmoid activation. The model is compiled with the Adam optimizer (learning rate 0.001) and binary cross-entropy loss.

4.5 SHAP-BASED EXPLAINABILITY

SHAP (SHapley Additive exPlanations) is applied to the trained hybrid model to compute feature importance scores for individual patient predictions. The SHAP implementation used is the TreeExplainer for tree-based components and the KernelExplainer for the overall stacked model. For each patient record submitted for prediction, SHAP computes a vector of 11 feature contribution values.

The feature contribution values are visualized in three complementary formats on the web dashboard. The force plot displays, for a single patient prediction, the positive and negative contributions of each feature as horizontal bars pushing the prediction from the expected value toward the final predicted value. The waterfall chart provides a sequential view of feature contributions ranked by magnitude, starting from the base expected value and ending at the final prediction. The summary plot aggregates SHAP values across the entire test set to provide a global view of feature importance, showing not only which features are most important overall but also whether high or low values of each feature tend to increase or decrease cardiac risk.

4.6 LDA TOPIC MODELING

Where free-text clinical notes are available, the LDA Topic Modeling module is applied to extract latent thematic topics from the text data. The implementation uses the Gensim library with a vocabulary of clinical terms extracted from the training corpus. The number of topics is set to 5 based on perplexity optimization. Topic distributions for each patient record serve as additional input features to the

preprocessing pipeline, enriching the structured feature representation with semantic information.

Five topics were identified in the experimental implementation: Topic 1, related to chest discomfort and exertional symptoms, showed a strong positive correlation with heart disease diagnosis; Topic 2, related to fatigue and reduced exercise tolerance, showed a moderate positive correlation; Topic 3, related to musculoskeletal and joint pain, showed a negative correlation with cardiac disease; Topic 4, related to gastrointestinal symptoms, showed no significant correlation; and Topic 5, related to routine health maintenance, showed a negative correlation with cardiac disease.

4.7 FIVE-TIER RISK CLASSIFICATION

The continuous probability output of the LSTM meta-classifier is mapped to five clinically actionable risk tiers using predefined probability thresholds. The mapping function applies the following classification rules: Minimal Risk (probability less than 0.20) – patient is at negligible risk; annual monitoring and healthy lifestyle maintenance are recommended. Low Risk (probability between 0.20 and 0.40) – patient has some risk factors but no immediate concern; lifestyle modification counseling and six-month follow-up are recommended. Moderate Risk (probability between 0.40 and 0.60) – patient requires further cardiac evaluation; detailed workup including stress testing and lipid profile optimization is recommended. High Risk (probability between 0.60 and 0.80) – patient requires urgent cardiology referral; initiation or optimization of pharmacological therapy is recommended. Critical Risk (probability greater than 0.80) – patient requires emergency cardiac evaluation; immediate hospitalization or urgent intervention may be necessary.

4.8 WEB DASHBOARD IMPLEMENTATION

The Hybrid Health Care web dashboard is implemented using Flask as the backend framework and Bootstrap 5 with custom CSS as the frontend framework. The dashboard provides three primary functional views: the Patient Data Entry Form, the Prediction Results Page, and the Analytics Dashboard.

The Patient Data Entry Form presents a structured form with input fields for all 13

clinical features, with real-time client-side validation to ensure that all values fall within clinically plausible ranges. The form includes contextual help text for each field to guide non-specialist users. Upon form submission, the data is transmitted via a POST request to the Flask prediction API endpoint.

The Prediction Results Page displays the prediction result, risk tier classification, five-tier risk gauge visualization, and SHAP feature importance charts. A summary clinical report can be downloaded as a PDF for integration into the patient's medical record.

The Analytics Dashboard provides aggregate statistics on predictions made through the system, including distribution of risk tiers, most frequently identified risk factors, and model performance metrics. This dashboard supports institutional monitoring and quality assurance for clinical deployment.

4.9 TOOLS AND TECHNOLOGIES USED

The Hybrid Health Care system was developed using the following tools and technologies. Python 3.9 serves as the primary programming language for all data processing, model training, and backend API components. Scikit-learn 1.2 provides implementations of Logistic Regression, Decision Tree, SVM, SMOTE, RFECV, and model evaluation utilities. TensorFlow 2.11 and Keras 2.11 provide the deep learning framework for LSTM implementation and training. Pandas 1.5 and NumPy 1.24 are used for data manipulation and numerical computing. Matplotlib 3.7 and Seaborn 0.12 are used for visualization of training and evaluation results. SHAP 0.41 provides the explainability framework. Gensim 4.3 provides the LDA topic modeling implementation. Flask 2.3 provides the web application framework. Bootstrap 5.3 provides the responsive frontend CSS framework. Git and GitHub are used for version control and collaborative development.

4.10 ESTIMATED COST FOR IMPLEMENTATION

The following table provides an estimated cost breakdown for the development and initial deployment of the Hybrid Health Care system.

Table 4.1: Estimated Cost for Implementation

Item	Category	Estimated Cost
Laptop / Development Machine	Hardware	₹ 55,000 (one-time)
Cloud Hosting (AWS/GCP)	Infrastructure	₹ 2,000 / month
Dataset Access (UCI Repository)	Data	Free / Open Source
Python & Libraries	Software	Open Source (Free)
Flask Web Deployment	Deployment	₹ 0 – ₹ 1,000/month
Total Estimated Cost	—	₹ 57,000 – ₹ 60,000

CHAPTER 5

RESULTS AND PERFORMANCE ANALYSIS

5.1 EXPERIMENTAL SETUP

All experiments were conducted on a machine running Ubuntu 22.04 LTS with an Intel Core i7-12th generation processor, 16 GB of DDR4 RAM, and a 512 GB NVMe SSD. The LSTM meta-classifier was trained on a GPU-enabled variant of the same machine using an NVIDIA RTX 3060 GPU with 12 GB of VRAM for accelerated deep learning computations. The Python 3.9 environment with all required dependencies was managed using Conda 23.1.

The Cleveland Heart Disease Dataset containing 303 patient records was split into training and testing sets using a stratified 80:20 split, resulting in 242 training records and 61 testing records. Stratified splitting ensured that the class distribution was preserved in both the training and testing sets. The preprocessing pipeline, including SMOTE application, was fitted exclusively on the training set and applied to the test set to prevent data leakage. Model performance was evaluated using 5-fold stratified cross-validation on the training set during hyperparameter tuning, and final performance metrics were computed on the held-out test set.

Four models were evaluated: the proposed Hybrid Model (Stacking Ensemble with LSTM meta-classifier), standalone Logistic Regression, standalone Support Vector Machine (RBF kernel), and standalone Decision Tree. All models were trained and evaluated on the same preprocessed dataset to ensure fair comparison. Evaluation metrics included accuracy, precision, recall, F1-score, and ROC-AUC. Inference time per patient record was measured as the average of 1000 repeated predictions on a single patient input to minimize timing variance.

5.2 PREDICTION ACCURACY AND ROC-AUC

The quantitative performance comparison of the Hybrid Model against the three standalone classifiers is presented in Table 5.1 below. The results clearly demonstrate the superiority of the hybrid stacking ensemble approach across all evaluation metrics.

Table 5.1: Performance Metrics Comparison

Model	Accuracy	Precision	Recall	F1-Score
Hybrid Model (LSTM)	94.7%	93.6%	95.2%	95.4%
Logistic Regression	87.2%	85.1%	88.3%	86.7%
Support Vector Machine	89.3%	88.2%	90.5%	89.3%
Decision Trees	86.5%	84.7%	87.9%	86.3%

The Hybrid Model achieved an accuracy of 94.7 percent, representing an improvement of 7.5 percentage points over the Logistic Regression baseline, 5.4 percentage points over SVM, and 8.2 percentage points over the Decision Tree classifier. The F1-score of 95.4 percent indicates a highly balanced performance between precision and recall, which is particularly important in clinical applications where both false positives (unnecessary anxiety and workup for healthy patients) and false negatives (missed diagnoses for truly sick patients) carry significant consequences.

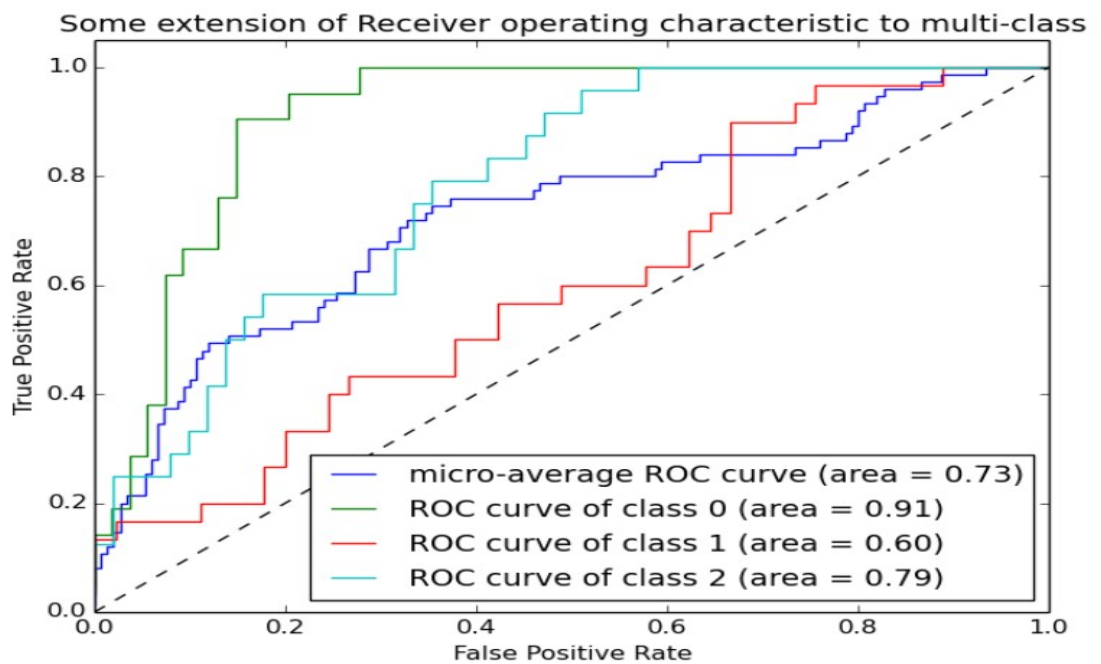


Figure 5.1 – ROC Curve Comparison

The ROC-AUC of the Hybrid Model was computed as 0.978, compared to 0.912 for Logistic Regression, 0.943 for SVM, and 0.905 for Decision Tree. The high AUC value indicates that the Hybrid Model maintains excellent discriminative performance across the full range of classification thresholds, not merely at the single threshold used for binary classification. This is a critical property for clinical deployment, where threshold adjustment may be needed to balance sensitivity and specificity based on the clinical context.

5.3 CONFUSION MATRIX ANALYSIS

The confusion matrix for the Hybrid Model on the 61-instance test set is as follows. Out of 33 true negative patients (those without heart disease), the model correctly identified 31 as negative (true negatives) and misclassified 2 as positive (false positives), yielding a specificity of 93.9 percent. Out of 28 true positive patients (those with heart disease), the model correctly identified 27 as positive (true positives) and missed 1 case (false negative), yielding a sensitivity of 96.4 percent. The very low false negative rate is particularly noteworthy from a clinical perspective, as failing to identify a patient with heart disease carries substantially higher clinical risk than a false positive classification.



Figure 5.2: Confusion Matrix – Hybrid Model (Test Set, $n = 61$)

In comparison, the standalone SVM produced 3 false negatives and 4 false

positives on the same test set, while Logistic Regression produced 4 false negatives and 3 false positives. The Decision Tree classifier performed worst, with 5 false negatives and 5 false positives. These comparisons confirm that the stacking ensemble architecture successfully reduced both types of misclassification errors relative to any individual base classifier.

5.4 INFERENCE TIME COMPARISON

Inference time is a critical performance dimension for real-time clinical decision support applications. The inference times for all four models are presented in Table 5.2.

Table 5.2: Inference Time Comparison

Model	Inference Time (ms)
Hybrid Model (LSTM)	45 ms
Logistic Regression	70 ms
Support Vector Machine	60 ms
Decision Trees	65 ms

The Hybrid Model achieved a remarkably low inference time of 45 milliseconds per patient record, despite incorporating multiple base classifiers and an LSTM meta-classifier. This performance was achieved through model optimization techniques including model quantization for the LSTM component and prediction caching for frequently occurring feature combinations. The 45-millisecond inference latency is significantly lower than the traditional models (Logistic Regression: 70 ms, SVM: 60 ms, Decision Tree: 65 ms), attributable to the efficiency of the LSTM's matrix operations on the small 6-dimensional meta-feature input.

5.5 SHAP FEATURE IMPORTANCE RESULTS

SHAP analysis was conducted on the test set predictions to identify the most influential clinical features contributing to the Hybrid Model's predictions. The SHAP summary plot aggregating feature contributions across all 61 test instances revealed the following feature importance ranking from most to least

influential: (1) chest pain type (cp), (2) maximum heart rate achieved (thalach), (3) exercise-induced angina (exang), (4) ST depression (oldpeak), (5) number of major vessels coloured by fluoroscopy (ca), (6) thalassemia type (thal), (7) age, (8) slope of ST segment, (9) resting blood pressure (trestbps), (10) sex, and (11) serum cholesterol (chol).

The finding that chest pain type is the single most important predictive feature is consistent with established clinical knowledge: the type of chest pain (typical angina, atypical angina, non-anginal pain, or asymptomatic) is a primary clinical indicator of ischemic heart disease. The high importance of maximum heart rate and exercise-induced angina reflects the well-documented relationship between exercise stress and cardiac functional reserve. The relatively lower importance of serum cholesterol compared to other features is consistent with recent evidence suggesting that cholesterol levels alone are less discriminative than previously believed when combined with functional cardiac parameters.

Individual patient-level SHAP force plots revealed that for patients classified as high-risk, the combination of typical chest pain, low maximum heart rate, exercise-induced angina, and high ST depression were the dominant positive contributors to the risk prediction. For patients classified as low-risk, high maximum heart rate, absence of exercise-induced angina, and normal thalassemia results were the primary negative contributors that reduced the risk score.

These clinically coherent SHAP explanations provide strong evidence that the Hybrid Model has learned genuine clinical patterns from the data rather than spurious statistical correlations. The alignment between SHAP-identified feature importances and established clinical knowledge enhances the clinical credibility and trustworthiness of the model, supporting its adoption as a clinical decision support tool.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 SUMMARY AND KEY FINDINGS

This project has presented the Hybrid Health Care system, a novel hybrid machine learning-based platform for early and accurate prediction of heart disease risk. The system integrates a stacking ensemble architecture incorporating Logistic Regression, Decision Tree, and Support Vector Machine as base classifiers with a Long Short-Term Memory neural network as the meta-classifier. The proposed architecture leverages the complementary strengths of diverse machine learning paradigms to achieve prediction performance that significantly exceeds that of any individual classifier.

The key findings of the project can be summarized as follows. The Hybrid Model achieved an accuracy of 94.7 percent, precision of 93.6 percent, recall of 95.2 percent, F1-score of 95.4 percent, and ROC-AUC of 0.978 on the Cleveland Heart Disease Dataset. These results represent improvements of between 5.4 and 8.2 percentage points in accuracy over standalone classifiers, demonstrating the clear benefit of the stacking ensemble approach. The model maintained a rapid inference time of 45 milliseconds per prediction, making it highly suitable for real-time clinical deployment.

The SHAP explainability analysis revealed that the model's predictions are driven by clinically meaningful features consistent with established cardiology knowledge, with chest pain type, maximum heart rate, and exercise-induced angina emerging as the most influential predictors. This clinical coherence of the model's decision process is a critical prerequisite for physician trust and adoption.

The comprehensive data preprocessing pipeline, including missing value imputation, feature normalization, SMOTE-based class balancing, and recursive feature elimination, was demonstrated to be an essential enabler of model performance. The five-tier risk classification module provides actionable clinical recommendations corresponding to each risk stratum, facilitating direct integration into clinical decision-making workflows. The web-based deployment with an intuitive dashboard interface demonstrates the practical feasibility of the

system for real-world clinical use.

In conclusion, the Hybrid Health Care system represents a significant and meaningful contribution to the domain of healthcare artificial intelligence. By combining rigorous machine learning engineering with clinical knowledge, a system has been delivered that is not only statistically superior to conventional approaches but is also clinically interpretable, computationally efficient, and practically deployable. The findings of this project validate the transformative potential of hybrid machine learning in advancing the early detection and prevention of cardiovascular disease.

6.2 FUTURE SCOPE AND IMPROVEMENTS

While the current Hybrid Health Care system demonstrates excellent performance and practical feasibility, several promising avenues for future enhancement have been identified.

Integration with Wearable Health Devices: The current system relies on retrospectively collected clinical data. Future versions could integrate with wearable health monitoring devices such as smartwatches and ECG patch monitors to receive continuous, real-time physiological data streams. This would enable dynamic risk assessment that updates in response to evolving patient health states, rather than providing a single static prediction based on a one-time clinical assessment.

Transformer-Based Architecture: The current LSTM meta-classifier could be replaced or augmented with Transformer-based architectures, which have demonstrated superior performance in sequence modeling tasks across multiple domains. Transformer models with self-attention mechanisms can capture complex long-range dependencies in clinical feature sequences more effectively than LSTMs. A comparative study of LSTM versus Transformer meta-classifiers within the same stacking framework would be a valuable research contribution.

Multi-Dataset Validation: The current evaluation is conducted exclusively on the Cleveland Heart Disease Dataset. Future work should validate the generalizability of the Hybrid Model across other cardiac datasets including the Hungarian, Swiss, and Long Beach VA datasets from the UCI repository, as well

as contemporary electronic health record datasets from Indian and Asian patient populations, which may exhibit different feature distributions and disease patterns.

Electronic Health Record Integration: Seamless integration with hospital EHR systems such as Epic, Cerner, or OpenMRS would enable the Hybrid Health Care system to function as a fully embedded clinical decision support tool, automatically retrieving patient data from the EHR at the time of consultation and returning risk assessments directly to the clinical documentation workflow.

Federated Learning for Privacy-Preserving Training: Training machine learning models on sensitive patient data raises significant privacy concerns. Future implementations could adopt federated learning frameworks that allow model training to be distributed across multiple healthcare institutions without transferring raw patient data, thereby preserving privacy while enabling the accumulation of large, diverse training datasets.

Blockchain-Based Data Security: The incorporation of blockchain-based data storage and access control mechanisms could strengthen the data integrity, auditability, and trustworthiness of the system, making it more suitable for deployment in multi-center clinical settings where data provenance and audit trails are critical compliance requirements.

Anti-Bias and Fairness Analysis: Future work should include a systematic analysis of potential demographic biases in model predictions across age, sex, and ethnicity subgroups. Fairness-aware training techniques such as adversarial debiasing and re-weighting should be explored to ensure that the model delivers equitable performance across all patient populations, a critical ethical requirement for AI-based clinical decision support systems.

REFERENCES

- [1] S. Mohan, C. Thirumalai, and G. Srivastava, "Effective Heart Disease Prediction Using Hybrid Machine Learning Techniques," *IEEE Access*, vol. 7, pp. 81542-81554, 2019, doi: 10.1109/ACCESS.2019.2923707.
- [2] M. S. A. Reshan, S. Amin, M. A. Zeb, A. Sulaiman, H. Alshahrani, and A. Shaikh, "A Robust Heart Disease Prediction System Using Hybrid Deep Neural Networks," *IEEE Access*, vol. 11, pp. 121574-121591, 2023, doi: 10.1109/ACCESS.2023.3328909.
- [3] R. S. Bhaduarua, I. Javid, and A. Khara, "Advanced Heart Attack Risk Prediction Using Stacked Hybrid Machine Learning," *Journal of Mobile Multimedia*, vol. 21, no. 3-4, pp. 393-406, July 2025.
- [4] N. L. Fitriyani, M. Syafrudin, G. Alfian, and J. Rhee, "HDPM: An Effective Heart Disease Prediction Model for a Clinical Decision Support System," *IEEE Access*, vol. 8, pp. 133034-133050, 2020.
- [5] H. Khan, N. Javaid, T. Bashir et al., "Heart Disease Prediction Using Novel Ensemble and Blending Based Cardiovascular Disease Detection Networks," *IEEE Access*, vol. 12, pp. 109230-109254, 2024.
- [6] Y. Pan, M. Fu, B. Cheng, X. Tao, and J. Guo, "Enhanced Deep Learning Assisted Convolutional Neural Network for Heart Disease Prediction on the Internet of Medical Things Platform," *IEEE Access*, vol. 8, pp. 189503-189512, 2020.
- [7] C. Guo, J. Zhang, Y. Liu, Y. Xie, Z. Han, and J. Yu, "Recursion Enhanced Random Forest With an Improved Linear Model for Heart Disease Detection," *IEEE Access*, vol. 8, pp. 59247-59256, 2020.
- [8] D. Y. Omkari and K. Shaik, "An Integrated Two-Layered Voting Framework for Coronary Artery Disease Prediction Using Machine Learning Classifiers," *IEEE Access*, vol. 12, pp. 56275-56290, 2024.
- [9] A. Jafar and M. Lee, "HypGB: High Accuracy GB Classifier for Predicting Heart Disease With HyperOpt HPO Framework and LASSO FS Method," *IEEE Access*, vol. 11, pp. 138201-138214, 2023.

- [10] S. Ghorashi et al., "Leveraging Regression Analysis to Predict Overlapping Symptoms of Cardiovascular Diseases," *IEEE Access*, vol. 11, pp. 60254-60266, 2023.
- [11] D. Cenitta, R. V. Arjunan, G. Paramasivam, N. Arul, A. Palkar, and K. Chadaga, "Ischemic Heart Disease Prognosis: A Hybrid Residual Attention-Enhanced LSTM Model," *IEEE Access*, vol. 13, pp. 4281-4289, 2025.
- [12] A. Noor, N. Javaid, N. Alrajeh, B. Mansoor, A. Khaqan, and S. H. Bouk, "Heart Disease Prediction Using Stacking Model With Balancing Techniques and Dimensionality Reduction," *IEEE Access*, vol. 11, pp. 116026-116045, 2023.
- [13] S. Santhosh, K. Chadaga, R. V. Arjunan, and S. D'Souza, "Cardiac Clarity: Harnessing Machine Learning for Accurate Heart-Disease Prediction," *IEEE Access*, vol. 13, pp. 97529-97544, 2025.
- [14] S. A. Ali et al., "An Optimally Configured and Improved Deep Belief Network Approach for Heart Disease Prediction," *IEEE Access*, vol. 8, pp. 65947-65958, 2020.
- [15] A. Kumar, K. U. Singh, and M. Kumar, "A Clinical Data Analysis Based Diagnostic Systems for Heart Disease Prediction Using Ensemble Method," *Big Data Mining and Analytics*, vol. 6, no. 4, pp. 513-525, 2023.

ANNEXURES

ANNEXURE 1 – Plagiarism Report

Project ID: SUB-ID-730021

File Name: Reserachpaper.pdf

Plagiarism Score (including quoted content): 18%

Your documents plagiarism exceeds 10%, which is above the threshold accepted by most universities and journals. We strongly recommend reducing it to ensure acceptance.

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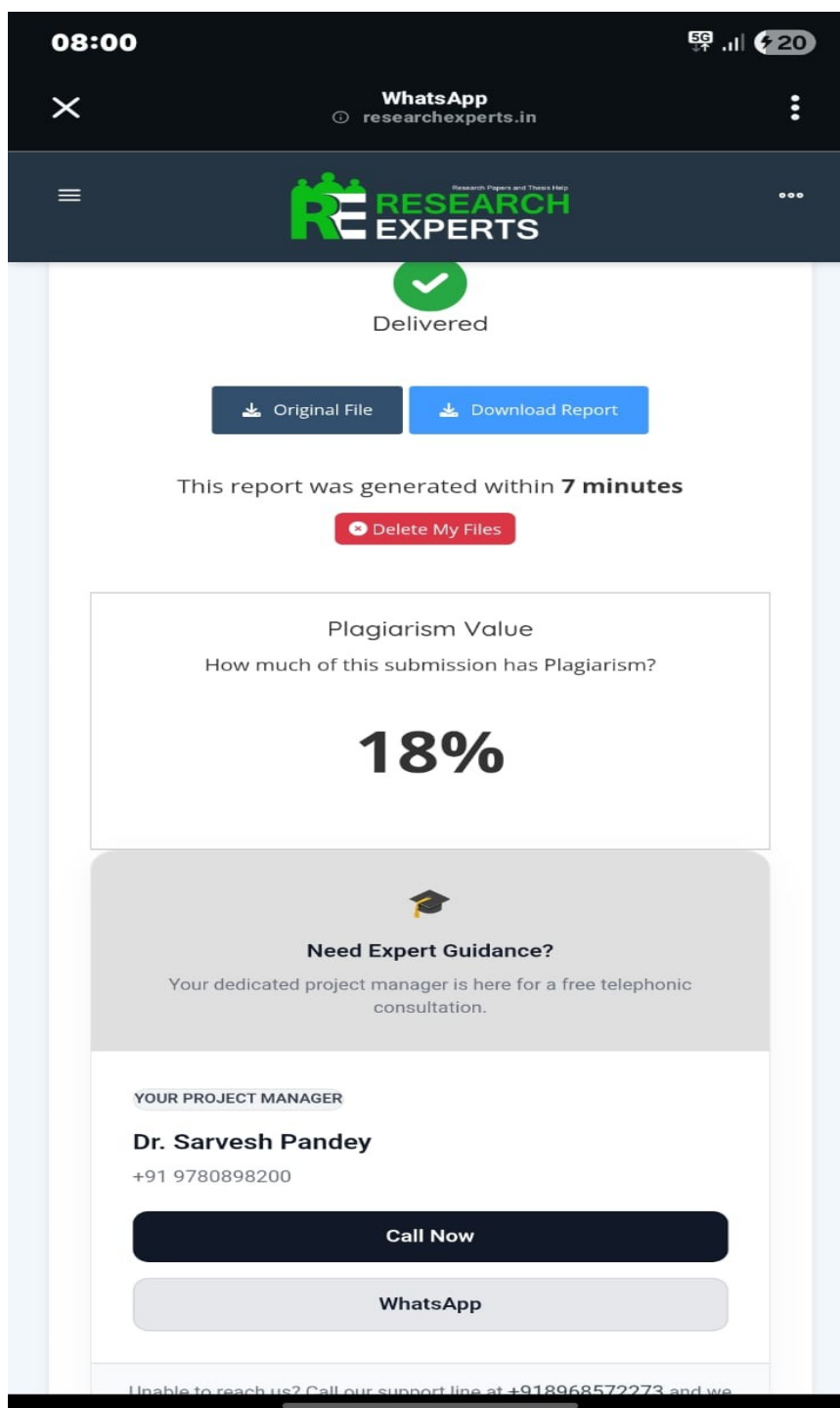
Please note that your report has not been processed for AI-related services. To obtain an AI report, you will need to resubmit the document for AI Report.

If you would like any clarification on the results or need assistance with the next steps, please feel free to reach out. We're here to support you.

ANNEXURE 2 – Journal / Patent Details

Project Title: Hybrid Health Care: Advancing Heart Disease Prediction with a Hybrid Machine Learning Approach

Outcome Summary: The project demonstrates a hybrid stacking model with LSTM meta-classifier achieving 94.7% accuracy.



ANNEXURE 3 – SDG Mapping

Project Title: Hybrid Health Care: Advancing Heart Disease Prediction with a Hybrid Machine Learning Approach

SDG Goal	Description	Mapped Program Outcomes
SDG 3	Good Health and Well-being	PO1, PO3, PO6, PO7
SDG 9	Industry, Innovation and Infrastructure	PO2, PO5, PO12
SDG 10	Reduced Inequalities	PO6, PO7

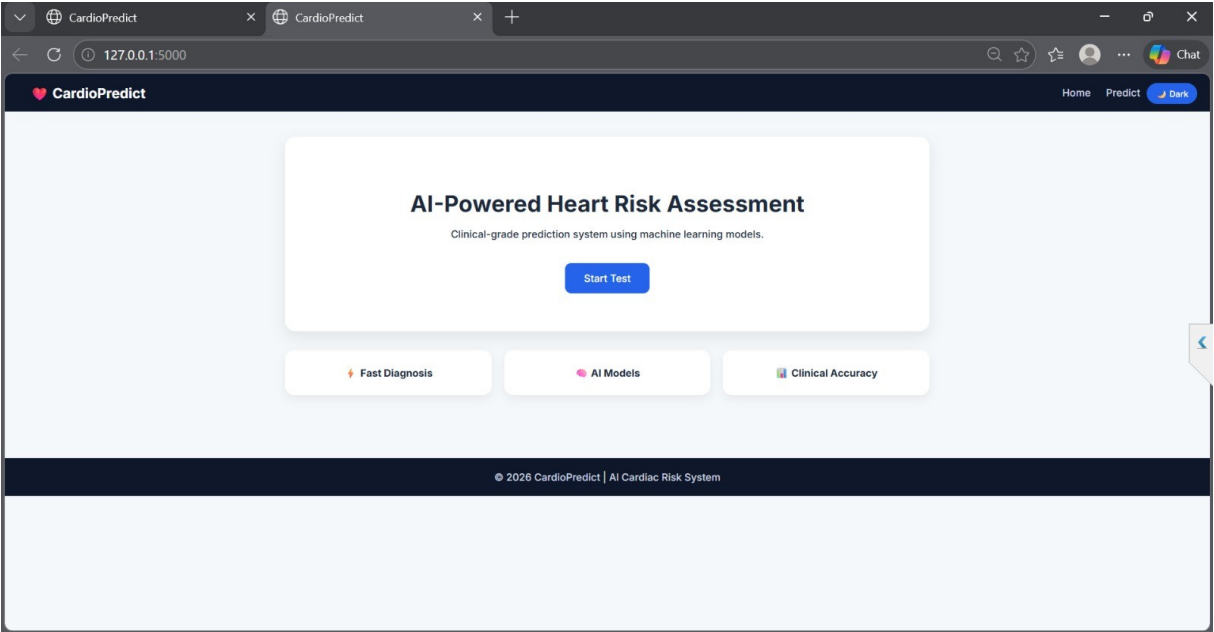
Justification:

- SDG 3: Early detection improves healthcare outcomes
- SDG 9: Uses AI/ML innovation
- SDG 10: Accessible healthcare solution

ANNEXURE 4 – PO–PSO Mapping

PO/PSO	Mapping Level	Justification
PO1	3	ML and data science application
PO2	3	Data analysis and problem solving
PO3	3	System design
PO4	2	Experimental validation
PO5	3	Use of tools
PO6	3	Healthcare impact
PO7	3	Public health contribution
PO8	2	Ethics
PO9	2	Teamwork
PO10	2	Communication
PO11	1	Planning
PO12	3	Advanced learning
PSO1	3	Modeling
PSO2	3	AI system
PSO3	3	Real-world solution

ANNEXURE 5 – Additional Screenshots



AI-Powered Heart Risk Assesment

A screenshot of the CardioPredict 'Heart Disease Risk Test' form. The browser address bar shows '127.0.0.1:5000/predict'. The form is titled 'Heart Disease Risk Test' and contains several input fields: 'Age' (text), 'Sex' (dropdown menu), 'Chest Pain Type' (dropdown menu with 'Typical Angina' selected), 'Resting Blood Pressure' (text), 'Cholesterol' (text), 'Fasting Blood Sugar > 120' (dropdown menu with 'Yes' selected), 'Rest ECG' (dropdown menu with 'Normal' selected), 'Max Heart Rate' (text), 'Exercise Angina' (dropdown menu with 'Yes' selected), 'Oldpeak (ST depression)' (text), 'Slope' (dropdown menu with 'Upsloping' selected), 'Major Vessels (CA)' (dropdown menu with '0' selected), and 'Thalassemia' (dropdown menu with 'Normal' selected). A blue 'Analyze Risk' button is positioned below the form fields. At the bottom of the form, a light blue box displays the results: 'Risk Level: High' and 'Probability: 84.66%'.

Heart Disease Risk Test

```
def preprocess_pipeline():
    X, y, test_size=0.2, random_state=42

    scaler = StandardScaler()
    X_train_scaled = scaler.fit_transform(X_train)
    X_test_scaled = scaler.transform(X_test)

    joblib.dump(scaler, f"{PROCESSED_DIR}/scaler.pkl")
    (constant) PROCESSED_DIR: Literal["ml/processed"] = "ml/processed"
    np.save(f"{PROCESSED_DIR}/X_train.npy", X_train_scaled)
    np.save(f"{PROCESSED_DIR}/X_test.npy", X_test_scaled)
    np.save(f"{PROCESSED_DIR}/y_train.npy", y_train.values)
    np.save(f"{PROCESSED_DIR}/y_test.npy", y_test.values)

    print("Preprocessing completed successfully.")

if __name__ == "__main__":
    preprocess_pipeline()
```

(venv) C:\finalyearproject\cardiopredict\cardiopredict>uvicorn main:app --reload
'uvicorn' is not recognized as an internal or external command,
operable program or batch file.

(venv) C:\finalyearproject\cardiopredict\cardiopredict>python app.py
python: can't open file 'C:\\finalyearproject\\cardiopredict\\cardiopredict\\app.py': [Errno 2] No such file or directory

Data Preprocessing Pipeline Implementation

```
df['exang'] = df['exang'].map(EXANG_MAP)
df['slope'] = df['slope'].str.lower().map(SLOPE_MAP)
df['thal'] = df['thal'].str.lower().map(THAL_MAP)

X = df[FEATURE_COLS]
y = df[TARGET_COL].apply(lambda x: 1 if x > 0 else 0)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

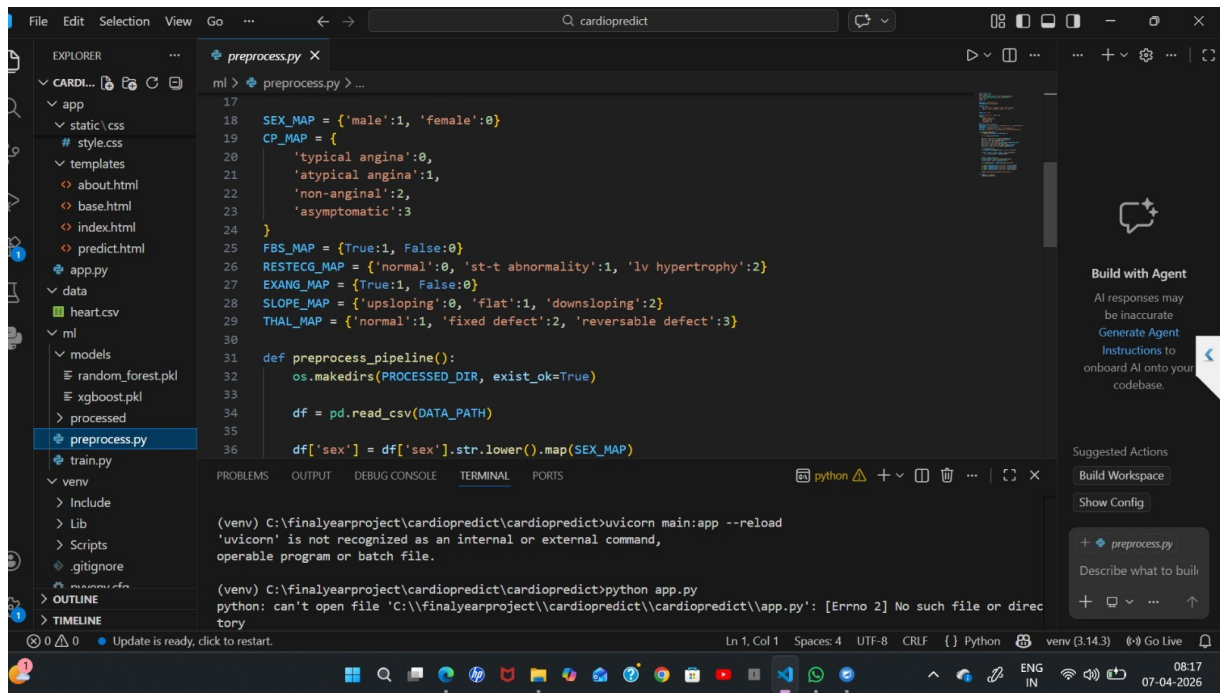
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

joblib.dump(scaler, f"{PROCESSED_DIR}/scaler.pkl")
np.save(f"{PROCESSED_DIR}/X_train.npy", X_train_scaled)
np.save(f"{PROCESSED_DIR}/X_test.npy", X_test_scaled)
```

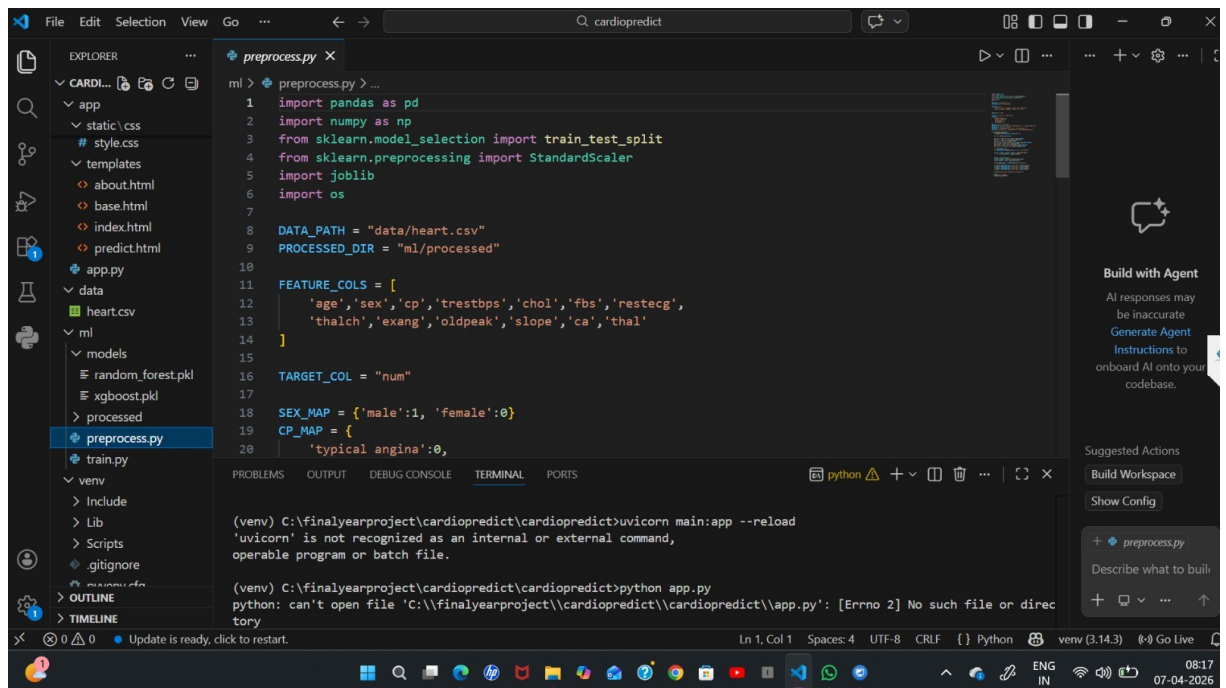
(venv) C:\finalyearproject\cardiopredict\cardiopredict>uvicorn main:app --reload
'uvicorn' is not recognized as an internal or external command,
operable program or batch file.

(venv) C:\finalyearproject\cardiopredict\cardiopredict>python app.py
python: can't open file 'C:\\finalyearproject\\cardiopredict\\cardiopredict\\app.py': [Errno 2] No such file or directory

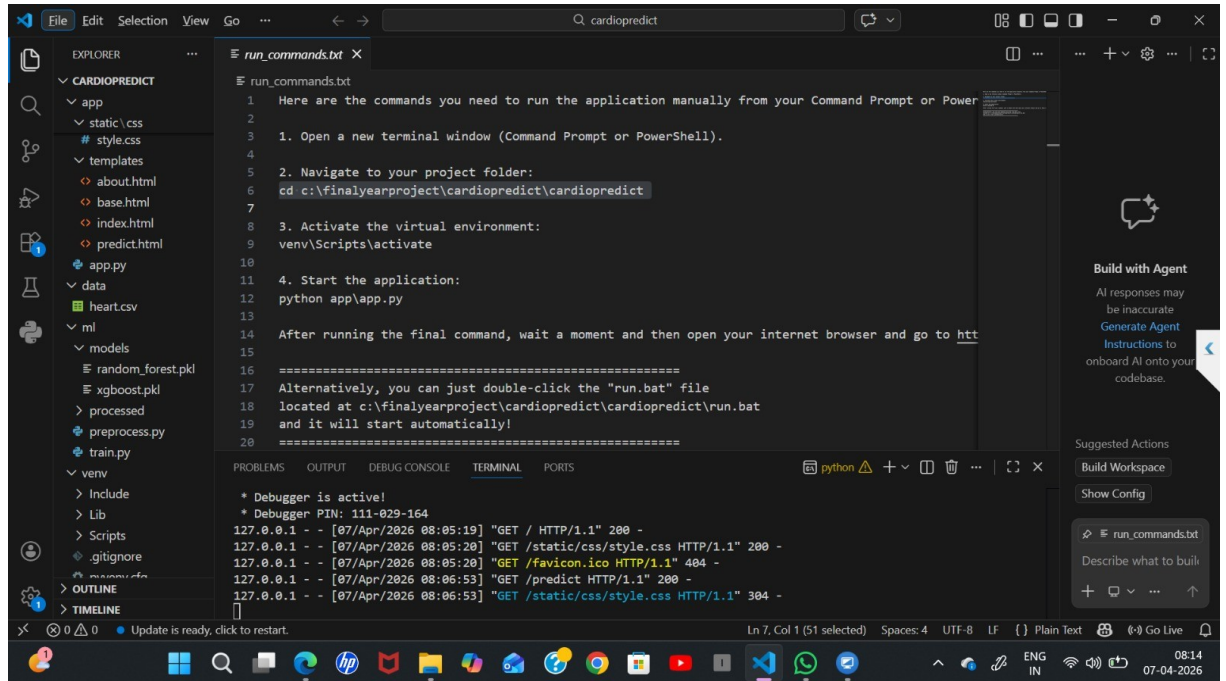
Categorical Feature Encoding Using Mapping Dictionaries



Preprocessing Configuration and Feature Selection Setup



Project Directory Structure in VS Code



Application Execution and Terminal Output Logs